

# PatientTriage.ai: Business Proposal & Enterprise Strategy

## Closing the Emergency Waiting-Room Surveillance Gap

Accenture Innovation Challenge 2026

Our idea in one sentence: *PatientTriage.ai helps Emergency Department staff notice which waiting patients may be getting worse and need attention first.*

### 1. The Business Problem

Emergency Departments are extremely busy. Triage helps doctors and nurses understand a patient's condition when they arrive, but triage is only a snapshot. A patient may then wait for 2.5–4.5 hours unmonitored. During that period, condition can change, vital signs can become abnormal, data can become old, or a monitor can disconnect. Staff may not immediately notice the change, creating a dangerous surveillance gap between triage and treatment. **PatientTriage.ai is designed to help hospitals manage this gap.**

### 2. Our Solution: Extra Safety Layer for the Waiting Room

PatientTriage.ai works like an extra safety layer for the waiting room. It continuously looks at available patient information and answers three simple questions:

- **1. Is the patient getting worse?** Checks changes in vital signs over time ( $\Delta SpO_2$ ,  $\Delta HR$ ).
- **2. Is the information still reliable?** Checks how old the latest observations are (shelf-life expiration).
- **3. Do we know enough about the patient?** If important information is missing, it asks staff to physically verify the patient (*Unknown  $\neq$  Safe*).

Then it creates a simple focused list: **Who needs attention first?**

### 3. Simple User Flow & Decision Architecture

Patient Arrives ■■■ Intake Triage ■■■ Waiting Lounge ■■■ Continuous Check ■■■ 4 States (■/■/■/■) ■■■ Priority Queue ■■■ Clinician Decides

*The system supports the clinician. It does not replace the clinician. Medical professionals retain 100% decision authority.*

### 4. Value for Hospital Staff & Leadership

| Role                | Current Problem                                            | PatientTriage.ai Value Delivery                                                           |
|---------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Triage Nurses (RNs) | Overwhelmed tracking 40+ waiting patients; alarm fatigue.  | Focused 'Next 5 Minutes' queue highlights patients needing immediate attention.           |
| Emergency Doctors   | Blind to which waiting patients deteriorated since intake. | Instant visibility into who has deteriorated, needs reassessment, or has warning signs.   |
| Hospital Leadership | Delayed diagnosis risks, sentinel waiting lounge events.   | Better waiting-room visibility, fewer missed deteriorations, better clinical audit trail. |
| Finance Teams       | LWBS walkout revenue losses, uncompensated ICU transfers.  | Captures modeled \$3.58M net annual value from recovered walkouts & fewer ICU crashes.    |

5. Business Impact & Value Breakdown (Modeled 500-Bed Facility)

Model based on 65,000 annual ED visits, 500 beds, and \$1,200 average baseline ED revenue per visit (projections from business model, not actual hospital results):

| Area of Value                                       | Expected Operational Impact                   | Modeled Annual Value |
|-----------------------------------------------------|-----------------------------------------------|----------------------|
| 1. Fewer patients leaving without being seen (LWBS) | 30% reduction via proactive re-engagement     | \$1,123,200 / yr     |
| 2. Fewer waiting-room ICU escalations               | 93 avoided stays (@ \$15k per stay)           | \$1,395,000 / yr     |
| 3. Lower malpractice liability risk                 | 40% reduction via deterministic audit logs    | \$480,000 / yr       |
| 4. Staff retention & overtime savings               | Lower nurse turnover + 15% overtime reduction | \$378,000 / yr       |
| 5. Better ED throughput & boarding                  | 30-minute improvement via optimized dispatch  | \$445,000 / yr       |
| Total Gross Annual Value                            | —                                             | \$3,821,200 / yr     |
| Modeled Software & Hardware Subscription            | Enterprise annual software license + support  | -\$240,000 / yr      |
| Net Estimated Annual Value (14.9x ROI)              | Net Return on Investment                      | +\$3,581,200 / yr    |

6. Hospital Integration, Safety Rules & Fail-Safe Design

- **HL7 FHIR Integration:** Works alongside existing Electronic Health Record (EHR) systems using standard data such as Patient Encounters and Vital Observations. Local edge processing avoids cloud AI latency.
- **Safety & Fixed Thresholds:** Serious conditions (SpO<sub>2</sub> < 85%, SBP < 75 mmHg) directly trigger an escalation without being masked by a statistical score.
- **Fail-Safe Behavior:** If network connectivity or telemetry fails, the system prompts staff to perform manual checks while local safety alerts remain active.

7. Target Market & Implementation Roadmap

- **Target Market:** 5,500+ Emergency Departments across US & Europe with high patient volumes and long wait times.
- **Stage 1 — Prototype (Completed - Q3 2026):** Built and tested with 51 automated tests passing.
- **Stage 2 — Shadow Deployment (Q4 2026):** Run alongside existing hospital processes to compare recommendations with clinician decisions.
- **Stage 3 — Pilot (Q1–Q2 2027):** Deploy in ED setting to measure time to escalation, wait times, LWBS, and alert usefulness.
- **Stage 4 — Scale (Q3 2027+):** Expand to more hospitals, ambulance telemetry, and regional patient flow coordination.

**Final Proposition:** PatientTriage.ai is not another triage system. It is a waiting-room monitoring and prioritization layer that works after triage and before treatment, answering: “Who has changed, and who needs attention now?”

Because a patient being stable at 2:00 PM does not mean they will still be stable at 4:00 PM. Triage is a snapshot. Risk isn't.